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AHA SCIENTIFIC STATEMENT

Infective Endocarditis: Diagnosis, Antibiotic Therapy, and Management: A Scientific Statement From the American Heart Association

Endorsed by the International Society for Cardiovascular Infectious Diseases (ISCVID) and the Canadian Cardiovascular Society (CCS)

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ABSTRACT: Since the publication of the 2015 American Heart Association scientific statement on infective endocarditis, its incidence has increased, its epidemiology has changed, and novel diagnostic and treatment options have become available. Updated case definitions, non–culture-based pathogen identification methods, and advanced multimodality imaging techniques have emerged. A randomized controlled trial of partial oral therapy published in 2019 may change the treatment landscape for the foreseeable future. Percutaneous mechanical aspiration of right-sided vegetations is being used in select patients with uncontrolled infection despite appropriate antimicrobial therapy. The importance of an endocarditis team has become increasingly recognized in the management of this life-threatening syndrome. Therefore, multidisciplinary experts in cardiovascular infections were commissioned to develop this updated scientific statement on the diagnosis and management of infective endocarditis.

Key Words: AHA Scientific Statements ■ cardiovascular infections ■ echocardiography ■ embolism ■ endocarditis ■ heart valves ■ treatment outcome

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Infective endocarditis (IE) remains a life-threatening infection with substantial short- and long-term risks for those who survive an initial episode. Interest in IE has grown substantially due to notable shifts in its epidemiology and rising incidence. Key contributors include an increasing incidence of injection drug use–related IE, especially in the United States; an aging population with medical comorbidities; a high prevalence of an array of indwelling cardiovascular devices; and bioprosthetic materials.[1] Advancements in both IE diagnosis and management have also fueled interest. The updated 2023 Duke/International Society for Cardiovascular Infectious Diseases (ISCVID) IE criteria incorporate novel diagnostic modalities, including molecular pathogen identification techniques (pathogen-specific, multiplex real-time and broad-range polymerase chain reaction and shotgun metagenomic sequencing) to identify causative pathogens and multimodality imaging. These advances have improved the diagnosis of IE, thereby altering management, and may be associated with improved outcomes, particularly when a multispecialty endocarditis team of healthcare clinicians with expertise in IE is involved in decision-making and management (Figure).

A paradox remains: mortality rates associated with IE remain persistently high. Randomized controlled trials (RCTs), the cornerstone of evidence generation, are rare, and IE management has historically relied on observational studies and expert opinion. Only 1 RCT focused on antimicrobial therapy in patients with IE with adequate cohort size and follow-up has been conducted.[2]

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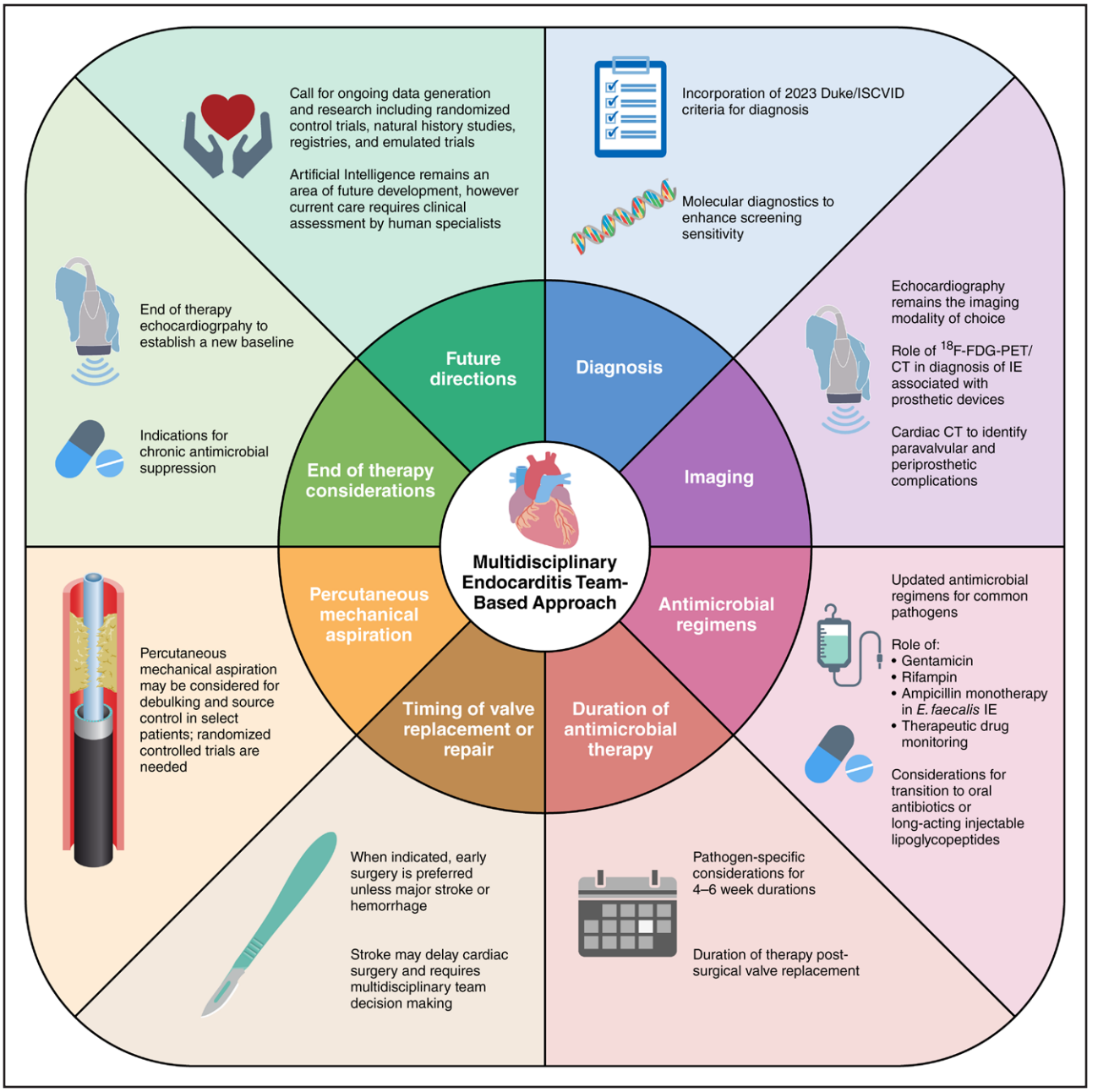


Figure. Multidisciplinary approach to infective endocarditis care.
CT indicates computed tomography; FDG-PET, fluorodeoxyglucose positron emission tomography; IE, infective endocarditis; and ISCVID, International Society for Cardiovascular Infectious Diseases.

This study, which evaluated transitional intravenous to oral antibiotic therapy, was published after the 2015 American Heart Association (AHA) scientific statement.[3] There are several factors related to this lack of RCT data, including the relative rarity of hospital admissions for IE, the inherent challenges of recruiting and obtaining informed consent from patients during the acute phase of a life-threatening illness, and the high cost, large scale, and prolonged duration of strategy trials. Persistent clinician and institutional biases regarding patient evaluation and treatment approaches also play a role, alongside limited industry motivation given the relatively small market size and modest financial return anticipated in an IE trial.

Unlike the 2015 AHA scientific statement and its predecessors, more recent AHA scientific statements do not include formal diagnostic and treatment recommendations. The current writing group has been tasked with providing management suggestions despite limited RCT evidence. Members of an international writing group with longstanding expertise in IE were invited to develop an updated scientific statement to support front-line clinicians managing complex cases that often fall outside

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of existing guidelines. Consensus was sought, but not mandatory.

To further assist clinicians, 4 recently published strategic IE-focused AHA scientific statements offer focused guidance on specific IE-related topics: IE prevention (2021),[4] IE in people who inject drugs (PWID; 2022),[1] cardiac implantable electronic device (CIED) infections (2024),[5] and blood culture–negative endocarditis (2025).[6] There is also an American College of Cardiology/AHA clinical guideline for the management of patients with valvular heart disease that includes a section on IE (2020).[7] In addition, there have been 3 AHA science advisories that address the risk of IE associated with nondental invasive procedures (2023),[8] selected nonvalvular cardiovascular device infections (2026),[9] and percutaneous mechanical aspiration (PMA) in right-sided IE (2026).[10]

Given the clinical variability and complexity of IE, the guidance provided herein is intended to support, rather than supplant, decisions in individual patient management, ideally made in collaboration with a multidisciplinary endocarditis team.

METHODS

A review process by the 12 writing group members that included identification, acceptance, and approval by the AHA manuscript oversight committee was initially conducted. A start-up call that included writing group members and AHA administrative staff was held on June 26, 2025. An EndNote library of publications was provided by a Mayo Clinic librarian. Screened publications were limited to English language and were identified by an Embase and Medline search on June 27, 2025. The search terms and format are listed in the Supplemental Material (Supplemental Table 1). Each writing group member had access to the list of publications that could be used as preparation for the development of this scientific statement. The writing group also provided clinical vignettes on complex IE diagnosis and management perspectives (Supplemental Vignettes).

ENDOCARDITIS TEAM

The Endocarditis Team and the Diagnosis, Treatment, and Outcomes of IE

The endocarditis team has been embraced globally as the cog for individual patient care (Figure). Given the complexity of IE and its often poor outcomes, the writing group endorses a comprehensive multispecialty team with expertise in the diagnosis and management of IE. The endocarditis team typically includes cardiology with expertise in imaging, cardiovascular surgery, infectious diseases, clinical microbiology, neurology, primary care, radiology, dentistry, pharmacy, and nursing. Addiction medicine expertise is essential in the management of IE in PWID and is also included in the endocarditis team. In a systemic review and meta-analysis, dedicated multidisciplinary teams for IE management were associated with improved short-term mortality, decreased time to surgery, and an increased rate of surgery. This team-based approach also enables standardization of diagnosis and management of patients with IE, provision of support to frontline clinicians at community hospitals in a hub-and-spoke model, and educational opportunities.

Key Take-Home Points

- The endocarditis team model is optimal for individualized patient care.
- Endocarditis team involvement has been associated with improved outcomes.

DIAGNOSIS OF IE

Changes in the 2023 Duke/ISCVID Criteria Versus the 2000 Modified Duke Criteria

In response to emerging shifts in microbiology, risk factors, and diagnostics of IE, the 2023 Duke–ISCVID IE criteria introduced several key updates to the 2000 modified Duke criteria.[11] These key updates included incorporating new molecular diagnostic techniques into the pathologic criteria for definite IE, expanding the list of typical IE pathogens, and adding situationally typical organisms associated with IE involving prosthetic material, new modalities to imaging criteria, and new major surgical criteria for operative inspection of the infected valve (Supplemental Table 2[11] and Supplemental Table 3[11]). In the validation studies that followed, the 2023 Duke/ISCVID criteria were shown to be more sensitive than the 2000 modified Duke criteria[12] and, depending upon the definition used, had either comparable[13,14] or modestly reduced specificity.[12,15]

Key Take-Home Points

- The Duke criteria have evolved to improve the clinical diagnosis of IE.
- The 2023 Duke/ISCVID criteria are used for the contemporary diagnosis of IE.
- The addition of new molecular diagnostics and imaging modalities has enhanced the sensitivity of the 2023 Duke/ISCVID criteria compared with that of the 2000 modified Duke criteria with a null to modest reduction in specificity.

The 2023 Duke–ISCVID Criteria Versus the 2023 ESC Diagnostic Criteria

The 2023 European Society of Cardiology (ESC) diagnostic criteria[16] and the 2023 Duke–ISCVID criteria[11]

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both introduced updates to the 2000 modified Duke criteria for diagnosing IE. There are subtle yet important differences between these criteria. They have similar sensitivity, but the 2023 Duke–ISCVID criteria have higher specificity.[13] The 2023 ESC diagnostic criteria included valve thickening as a distinctive echocardiographic feature and the incorporation of hematogenous osteoarticular complications in the vascular phenomena criteria.

Enterococcus species are recognized as important pathogens by both documents, especially in older patients with prosthetic devices. Both 2023 documents updated the list of typical IE pathogens to include *Enterococcus faecalis*, moving beyond the previous requirement of "community-acquired without a known primary focus." The 2023 Duke–ISCVID criteria demonstrate higher sensitivity (79%) for diagnosing IE in patients with enterococcal bacteremia compared with the 2023 ESC diagnostic criteria (74%).[17] However, this increased sensitivity comes at the expense of specificity (55% for 2023 Duke–ISCVID versus 69% for 2023 ESC), meaning the former may classify more non IE cases as "possible IE." The 2023 Duke–ISCVID criteria also included "typical" microorganisms associated with IE involving prosthetic material and "atypical" microorganisms that occasionally or rarely cause IE when isolated from ≥3 separate blood cultures (Supplement Table 3). Only the 2023 Duke–ISCVID criteria simplified blood culture collection requirements by eliminating specific timing as separate venipuncture sites.[11] A French study recently validated this approach.[18]

Both 2023 documents include advanced imaging modalities, such as cardiac computed tomography (CCT) and ^{18}F-fluorodeoxyglucose positron emission tomography/computed tomography (^{18}F-FDG-PET/CT), as major criteria, especially for cases involving prosthetic material. In addition, CCT or ^{18}F-FDG-PET/CT may provide valuable adjunctive information for preoperative planning. The 2023 ESC diagnostic criteria also specify white blood cell single-photon emission CT/CT (WBC SPECT/CT) scans in cases of prosthetic valve endocarditis (PVE) early after surgery, which the 2023 Duke–ISCVID criteria do not. These advanced imaging modalities can be complementary, particularly in prosthetic valve IE. Access to these imaging modalities is limited worldwide.

Key Take-Home Points

- The 2023 ESC diagnostic criteria specify WBC SPECT/CT scans in cases of PVE early after surgery, which the 2023 Duke–ISCVID criteria do not.
- The 2023 Duke–ISCVID diagnostic criteria simplified blood culture collection requirements by eliminating specific timing and separate venipuncture sites.

The Role of Echocardiography in the Initial Diagnosis of IE

Echocardiography remains the cornerstone and first-line imaging modality for the evaluation of suspected IE. The 2023 Duke–ISCVID criteria include echocardiographic findings—such as vegetations, valvular lesions (perforation or new clinically significant regurgitation), and perivalvar or periprosthetic complications (abscess, pseudoaneurysm, fistula, or prosthetic valve dehiscence)—as major diagnostic criteria. In the 2023 Duke–ISCVID criteria, echocardiographic minor criteria were eliminated, underscoring the importance of high-certainty echocardiographic evidence for diagnosis (Supplemental Table 3). Prompt echocardiographic imaging is desired when IE is suspected, as delays in diagnosis are associated with greater valvular destruction, higher rates of embolic events, and increased need for surgical intervention.[19]

Transthoracic echocardiography (TTE) is the initial screening tool, given its wide accessibility and noninvasive nature, but its sensitivity is limited. TTE maintains high specificity (>90%), but most studies have reported sensitivities in the range of 40% to 66%, which vary with imaging quality.[20] In some cases, TTE may be adequate for evaluation of right-sided IE. In contrast, transesophageal echocardiography (TEE) offers superior (>90%) sensitivity and specificity. TEE is important for detecting prosthetic valve involvement, abscesses, and other perivalvular complications, for which the sensitivity of TTE is low (20%–46%). The limited sensitivity of TTE is related to acoustic shadowing, which can also affect TEE, particularly at the anterior aspect of the aortic root. TEE sensitivity also outperforms CCT in detecting vegetations (94% versus 64%), valvular perforation (81% versus 41%), and paravalvular leaks (69% versus 44%), whereas CCT provides superior detection of pseudoaneurysms or abscesses (78% versus 69%), and is frequently complementary to echocardiography in complex cases. TTE and TEE are both operator-dependent and could lead to both false-positive and false-negative diagnoses, underlining the importance of heart teams and imaging experts in IE.

Therefore, TEE is performed in patients with prosthetic valves or CIEDs, even if TTE is negative or inconclusive. In patients with native valves, TTE may be sufficient to exclude IE in those with low pretest probability and high-quality imaging; TEE is indicated when the posttest probability remains intermediate or high.[21,22] The decision to proceed with TEE in patients with native valves is guided by associated risk factors (eg, previous IE, injection drug use, congenital heart disease), clinical presentation (eg, fever, embolic phenomena, new murmur), and microbiologic findings (eg, persistent bloodstream infection [BSI], especially due to *Staphylococcus aureus*), but overall, a low threshold is favored. Echocardiography (TTE or TEE) may be repeated within 1 week

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after initial normal or inconclusive echocardiography results if the suspicion for IE remains high to assess for evolutional changes.

Key Take-Home Points

- Echocardiography is the imaging modality of first choice in the diagnosis of IE.
- TEE is performed in patients with suspected or confirmed prosthetic valve IE, CIEDs, and in select cases of native valve IE predicated on pretest probability, quality of TTE imaging, and patient risk factors.

Echocardiography in Patients With BSI

Echocardiography is not needed in all cases of BSI. The decision to obtain echocardiography depends on multiple factors, such as clinical presentation (community-acquired), pretest probability, identification of an infection source, stroke, persistent BSI, and pathogen designation. Certain microorganisms carry sufficiently high IE risk to warrant echocardiography. To refine the use of echocardiography in patients with BSI, several risk scores based on specific pathogens (*S aureus*, enterococci, and streptococci) have been developed, but their review is beyond the scope of this scientific statement.

Key Take-Home Points

- The need for echocardiography in the setting of BSI without clinical evidence of valvular heart disease depends on the organism identified and the underlying risk factors for developing IE.
- Several risk scores have been developed to assist in determining the need for echocardiography in Gram-positive bacteremia.

Echocardiography Before Transition to Partial Oral Therapy

As demonstrated in the POET trial (Partial Oral Treatment of Endocarditis), the writing group supports obtaining a TEE in left-sided IE at the time of transition to partial oral therapy (POT).[2]

Imaging During and at the end of Antimicrobial Therapy

Echocardiography

Echocardiography is warranted at any time during hospitalization and after hospital discharge when there is a change in clinical status. Although practice patterns vary, TTE is usually performed after completion of antibiotic therapy to establish a new baseline of valvular morphology and function against which future comparisons can be made. This strategy may be useful because vegetation morphology often evolves during healing. On serial imaging, lesions may thicken, decrease in size, become more echogenic, or persist as residual vegetations on valvular or chordal structures. These findings need to be interpreted within the clinical context, because residual valvular thickening or echogenicity is nonspecific and would not, in isolation, be taken to confirm healed infection or exclude persistent or recurrent disease. Rarely, follow-up TEE is pursued if poor TTE acoustic windows are encountered, or if the initial diagnosis was only or better characterized on index TEE, as in the case of PVE managed conservatively. Involvement of the endocarditis team is essential for making individualized patient care decisions based on echocardiography results.

TTE is routinely performed 6 to 12 weeks after any type of valve surgery independent of the indications for operation. The frequency of longer-term echocardiographic imaging is dictated by the severity of a residual valve lesion, valvular insufficiency over time, the type of heart valve substitute (if used), and clinical course. In complex surgical interventions, a TEE at 3 months could be considered.

Key Take-Home Points

- Echocardiography is needed during hospitalization and after hospital discharge for a change in clinical status.
- Echocardiography at the end of antimicrobial therapy is reasonable to establish a new baseline using TTE and, in select cases, TEE.

Transcatheter Heart Valve Replacement and IE

Transcatheter heart valve–related IE, and in particular transcatheter aortic valve replacement (TAVR) IE, is typically reported as a distinct subset of PVE due to increasing numbers of annual TAVR procedures performed worldwide and because of the unique challenges confronting endocarditis team members posed by these patients, who are typically older and more medically complex.

The choice of TAVR for the index procedure to treat severe aortic stenosis is usually predicated on an intermediate or high surgical risk profile and age, but this is changing.

Although the incidence of TAVR IE is low (0.3–1.9 per 100 patient-years), and not different from that of IE after surgical aortic valve replacement, there are differences between the entities regarding risk factors, timing of infection, microbiology, and outcomes.[23] For example, the leading cause of TAVR IE is enterococci, in contrast to both native valve endocarditis (NVE) and other types of PVE. TAVR explantation for

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structural deterioration (75%–80% of cases) or IE (the remainder) is the fastest-growing field within the specialty of cardiac surgery.[24] The operation is technically complex and has a learning curve that may be associated with poor outcomes (eg, bleeding, stroke, prolonged length of stay, early and late mortality), but can be accomplished safely at experienced centers after careful patient selection and when time allows preoperative optimization.[25] Single-center and limited registry studies have not convincingly demonstrated a survival advantage for TAVR explant over conservative therapy for TAVR IE, although these observational series are subject to high risk of bias and confounders.[16,25–27] At this time, these patients are managed like other patients with PVE at experienced centers.

Difficulties in TAVR IE Diagnosis

TAVR IE can be difficult to diagnose.[28] Symptoms, including fever and chills, may be blunted due to advanced age or multiple comorbid conditions. A sustained BSI without an alternative source due to Gram-positive cocci in a patient with a previous TAVR warrants prompt evaluation for TAVR IE. The sensitivity of both TTE and TEE is diminished in patients with TAVR due to artifacts and shadowing caused by the devices' anchoring stents. This may prompt discussion by the endocarditis team to evaluate the pros and cons of other types of multimodality imaging (if available), including CCT angiography, ^{18}F-FDG-PET/CT, or WBC SPECT/CT.[27]

Key Take-Home Point

- Sensitivity of both TTE and TEE is diminished in TAVR IE, and multimodality imaging may be required in the diagnosis.

TRANSCATHETER PULMONARY VALVE REPLACEMENT

Transcatheter pulmonary valve replacement (TPVR) is much less commonly performed than TAVR, but TPVR IE deserves comment. A femoral approach is predominantly used for TPVR in children or young adults with congenital heart disease who need treatment of pulmonic valve dysfunction. The median time to IE is prolonged (2.7 years), unlike in TAVR IE, in which almost two-thirds of cases are seen within 1 year of device placement. This suggests that procedure-related factors (femoral access) of TPVR are less likely to be important in the pathogenesis of IE. Factors associated with TPVR IE include previous IE, younger age, and elevated residual gradient after valve implant. The microbiology of TPVR IE differs from TAVR IE in that TPVR has low rates of enterococci as a pathogen, despite a transfemoral approach. This observation is of keen interest when considering the epidemiology of TPVR IE, particularly given that that enterococcal IE is generally uncommon in younger individuals and is most frequently seen in older men, particularly those with gastrointestinal or genitourinary tract conditions. This raises the question of whether there will be a reduction in enterococcal IE cases among patients with TAVR as the shift in placement involves low- and intermediate-risk younger populations. In addition, TPVR IE frequently occurs in patients with complex congenital heart disease who have undergone ≥1 previous open-heart surgeries.

Regarding the diagnosis of TPVR IE, the low sensitivity of echocardiography is related to the challenging visualization of the right ventricular outflow tract and pulmonic valve, even with TEE.[29] Other limitations include the presence of other devices among patients with congenital heart disease and the likelihood of previous open surgical procedures, with a possible impact on the underlying cardiac anatomic features. Therefore, multimodality imaging, including intracardiac echocardiography with or without ^{18}F-FDG-PET/CT, CCT, or cardiac magnetic resonance imaging (MRI), may be needed to secure an IE diagnosis. The role of cardiac MRI in IE diagnosis is limited, but it may demonstrate perivalvular inflammation, conduit wall thickening, and right ventricular outflow tract obstruction severity.

Transcatheter repair or replacement of the mitral and tricuspid valves are not addressed herein, but future expectations include the ongoing expansion of transcatheter valve interventions with an ever-increasing variety of devices, all of which will harbor some risk of complicating IE as confirmed by evolving diagnostic modalities.

Key Take-Home Points

- Sensitivity of TTE and TEE is diminished in TPVR IE, and multimodality imaging may be required in the diagnosis.
- Echocardiographic expertise in congenital heart disease is favored.
- In contrast to TAVR IE, TPVR IE has lower rates of enterococci despite a transfemoral approach.

Advancements in Molecular Diagnostics to Enhance the Detection of Pathogens in IE

Over the past decade, molecular diagnostics have emerged as the most consequential technologic advancement in the diagnosis of IE. In response, the AHA published a scientific statement on blood culture–negative endocarditis in 2025 that included an algorithmic framework to guide frontline clinicians in the selection and interpretation of appropriate diagnostic tests (Supplemental Figure 1).[6]

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Obstacles to Effective Integration of Traditional and Molecular Methods for Improved Diagnostics

Access to advanced molecular diagnostic modalities remains limited in Western countries and even more so in low- and middle-income countries. Financial barriers including high test costs and inconsistent insurance coverage further restrict utilization.[30] To improve access and enable timely diagnosis of IE, broader insurance reimbursement and progressive cost reductions throughout technologic advancement are essential.

Selective Use of Advanced Microbiologic Modalities to Ensure Appropriateness and Cost-Effectiveness

Molecular testing is incorporated into the diagnostic workup of IE and is guided by clinical context and cost-effectiveness. If both blood and valve cultures are positive for a typical IE organism (eg, methicillin-susceptible *S aureus*), then molecular testing can be deferred. If possible, a preantibiotic plasma sample should be stored in the microbiology laboratory for possible future metagenomic sequencing if blood cultures are negative at 72 hours (Supplemental Figure 1). For typical pathogens, routine molecular testing rarely changes management when cultures are concordant.

If blood cultures grow an atypical organism for IE and subsequent valve cultures are negative at 72 hours, then it is reasonable to perform molecular testing of the collected valve specimen. For blood culture–negative endocarditis, molecular diagnostic testing can be used initially on valvular tissue, along with routine cultures and pathologic evaluation, to enable rapid identification of a causative organism.

Key Take-Home Point

- Molecular testing may be helpful when blood and valve tissue cultures are negative or inconclusive.

ADDITIONAL IMAGING MODALITIES IN IE

^{18}F-FDG-PET/CT

Over the past 2 decades, ^{18}F-FDG-PET/CT has become a valuable adjunctive imaging modality for the diagnosis and management of IE.[31–34] ^{18}F-FDG-PET/CT identifies areas of increased glucose metabolism, a hallmark of inflammation. FDG uptake may not reliably distinguish infection from sterile inflammation early after surgery; however, there are patterns that are more consistent with infection, underlying the importance of interpretation expertise. A key advantage of ^{18}F-FDG-PET/CT lies in its ability to combine functional and anatomic information, with the CCT (which includes ECG gating) component enabling precise localization of the structures corresponding to FDG uptake and thereby improving diagnostic accuracy.

Clinical Scenarios in Which ^{18}F-FDG-PET/CT Is Needed for IE Diagnosis

The integration of ^{18}F-FDG-PET/CT into the 2023 Duke/ISCVID criteria has substantially improved diagnostic sensitivity, particularly in PVE. ^{18}F-FDG-PET/CT can influence diagnosis by contributing either a major criterion (positive cardiac imaging for IE) or a minor criterion (evidence of vascular dissemination). Both the 2023 ESC diagnostic criteria and this scientific statement support ^{18}F-FDG-PET/CT use as a second-line adjunctive diagnostic tool for suspected PVE when echocardiography is negative or inconclusive but clinical suspicion remains high. Nonetheless, access to this modality remains limited in many centers.

The information provided by this imaging modality helps establish or refute an IE diagnosis; define the extent of infection (such as graft involvement) and sites of metastatic infection; enhance presurgical planning in nonurgent situations for refining risk stratification, facilitating procedural planning, and guiding timing; and monitor response to antimicrobial therapy. PVE studies have reported sensitivities of >80% to 90% with similar specificity (Supplemental Table 4[33]). The utility of ^{18}F-FDG-PET/CT seems similar for bioprosthetic and mechanical valves. In NVE, the sensitivity of ^{18}F-FDG-PET/CT is low, and its use is generally not recommended, although its high specificity may be of interest in select cases. By providing whole-body assessment, ^{18}F-FDG-PET/CT can identify extracardiac complications or infectious foci such as septic emboli, vertebral osteomyelitis, or splenic abscesses, which represent either secondary sites of infection (from embolic events or hematogenous dissemination) or the primary entry site. ^{18}F-FDG-PET/CT is less suited for detecting cerebral septic emboli or mycotic aneurysms due to high physiologic FDG brain uptake. In the absence of cardiac uptake, ^{18}F-FDG-PET/CT can also help identify alternative causes of sepsis. In patients managed conservatively, ^{18}F-FDG-PET/CT may be useful for monitoring disease progression, treatment response, and recurrence, although larger prospective studies are needed to confirm this notion.[35–37]

Proper patient preparation, including a high-fat, low-carbohydrate ketogenic diet for 24 hours before imaging, is essential to suppress physiologic myocardial uptake and improve specificity. In more urgent situations, appropriate images can be obtained without dietary preparation. Accurate interpretation requires expertise and experience; in particular, valvular, prosthetic, or perivalvular ^{18}F-FDG-PET/CT uptake patterns diagnostic for

IE are typically high-intensity, focal/multifocal or diffuse but heterogeneous, and persist on nonattenuation correction, in contrast to low-intensity, diffuse, and homogeneous nonspecific uptake. No standardized uptake thresholds have been defined, although the likelihood of infection increases with higher values. False-positives may occur due to recent surgery, sterile inflammation, AtriClip devices,[38] or foreign body reactions to substances such as BioGlue[39] that are used in complex procedures. Certain prosthetic valves, such as the Medtronic Mosaic valve, can also produce intense but diffuse and homogeneous uptake.[40] These issues underscore the importance of understanding the implanted material and surgical technique for accurate ^{18}F-FDG-PET/CT interpretation. False-negatives may result from low-grade infections, infections masked by previous antibiotic therapy, or inadequate suppression of myocardial uptake.[41–44] The use of ^{18}F-FDG-PET/CT in the early postoperative period (within 3 months after cardiac surgery) has traditionally been considered challenging, as FDG uptake from sterile postoperative inflammation could mimic infection and lead to false-positives. More recent data suggest that early postsurgical changes may be distinguishable from infection.[45]

Key Take-Home Points

- ^{18}F-FDG-PET/CT is a valuable adjunctive imaging modality in the diagnosis of PVE.
- ^{18}F-FDG-PET/CT is important in detecting extracardiac sites of infection.
- ^{18}F-FDG-PET/CT sensitivity is higher in CIED pocket infection compared with lead infection.
- Patient preparation is key when there is sufficient time, including a high-fat, low-carbohydrate ketogenic diet for 24 hours before imaging.

RADIOLABELED WBC SPECT/CT SCINTIGRAPHY

WBC SPECT/CT, like ^{18}F-FDG-PET/CT, integrates both metabolic and anatomic imaging by using autologous leukocytes labeled with either 99m–technetium-hexamethylpropyleneamine or 111-indium-oxine, and is more specific for infection, whereas ^{18}F-FDG-PET/CT reflects inflammation. This modality has long served as a valuable diagnostic tool in cases of suspected IE in which echocardiographic findings are inconclusive but clinical suspicion remains high.[46] WBC SPECT/CT is especially effective for identifying pyogenic infections involving cardiovascular prosthetic materials. These include prosthetic valve IE, CIED infections, ventricular assist devices, prosthetic vascular grafts, and early postoperative infections—clinical scenarios in which ^{18}F-FDG-PET/CT may have reduced specificity.[33]

Nonetheless, certain limitations are noteworthy. The sensitivity of WBC SPECT/CT is reduced in detecting small infectious foci or vegetations <1 cm, as well as in nonpyogenic or chronic infections caused by organisms such as *Coxiella burnetii*, *Tropheryma whipplei*, and fungi.[33]

The 2023 ESC diagnostic criteria recommended WBC SPECT/CT in patients with high clinical suspicion of prosthetic valve IE when echocardiography is negative or inconclusive and ^{18}F-FDG-PET/CT is unavailable.[16] The 2023 Duke/ISCVID criteria do not include WBC SPECT/CT findings as a major or a minor criterion. Despite its diagnostic utility, WBC SPECT/CT is less commonly used than ^{18}F-FDG-PET/CT, largely due to its lower sensitivity in NVE and the extended procedural time, which may require up to 24 hours to complete.[33]

Key Take-Home Points

- WBC SPECT/CT can be used in patients with high clinical suspicion of IE when echocardiography is inconclusive or negative and ^{18}F-FDG-PET/CT is unavailable.
- WBC SPECT/CT has lower sensitivity in NVE.

Indication for CCT

CCT is increasingly used in diagnosing and managing IE, including both NVE and PVE. In the 2023 ESC diagnostic criteria, CCT is recommended for detecting valvular lesions in cases of possible IE and for identifying paravalvular and periprosthetic complications when echocardiography is inconclusive.[16] This applies to both NVE and PVE. In the 2023 Duke–ISCVID criteria[11] and 2023 ESC diagnostic criteria, findings on CCT qualify as a major criterion. The role of CCT in the diagnosis of IE is also discussed in the 2020 American College of Cardiology/AHA guidelines for the management of valvular heart disease.[7]

Sensitivity and Specificity Differences Between CCT and TEE for Detecting Vegetations and Other Structural Complications

The primary indication of CCT is the evaluation of periprosthetic complications in PVE and perivalvular extension in NVE. TEE is limited by prosthetic shadowing and artifacts from mechanical discs, leaflets, and stent frames, especially on the anterior aspect.[47–49] Although TEE is generally more sensitive than CCT for detecting vegetations, particularly in NVE, in cases of PVE in some studies CCT has had a higher specificity (94% versus 74%).[47] CCT shows similar specificity to TEE for detecting fistulae (98% for both), with pooled sensitivities ranging from 79% to 98% for CCT. In addition, CCT can identify dehiscence of prosthetic valves, but this also requires multiphase imaging to detect a rocking motion.

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CCT in the Preoperative Assessment of Coronary Arteries

CCT is valuable in planning surgical treatment for IE. It can aid in the assessment of native coronary arteries as an alternative to invasive angiography in low- and intermediate-risk patients to exclude clinically significant proximal coronary artery disease. In those who have undergone previous coronary artery bypass grafting, CCT can be used to evaluate the patency and the course of the grafts. When assessing coronary artery disease, slower heart rates are beneficial, but achieving these can be difficult in critically unwell patients with hemodynamic instability or arrhythmia such as atrial fibrillation. Furthermore, caution is needed when using iodinated contrast agents in patients with renal dysfunction, which commonly coexists in those with sepsis.

Key Take-Home Points

- CCT is an essential tool in identifying paravalvular and periprosthetic complications when echocardiography results are inconclusive, especially in PVE.
- CCT is valuable in planning surgical treatment for IE and, with appropriate imaging protocols, can provide an assessment of native coronary arteries for coexistent disease and bypass grafts, as an alternative to invasive angiography in select patients.

Role of CMR in IE

CMR has a limited role in the evaluation of cardiac manifestations of IE compared with echocardiography and CCT.

Indications for Brain MRI

Neurologic involvement occurs in patients with IE and includes ischemic stroke, intracranial hemorrhage, brain abscess, and cerebral microbleeds. MRI offers superior sensitivity compared with CT and is the modality of choice for the diagnosis and evaluation of these complications. MRI is more sensitive than CT for detecting acute ischemic events, especially small cortical and subcortical infarcts caused by septic emboli. Diffusion-weighted imaging can identify acute ischemia even in the absence of clinical symptoms, enabling recognition of "silent" embolic events that would be a minor diagnostic criterion for definite IE.[50,51] MRI also detects brain abscesses, typically presenting as ring-enhancing lesions, and can identify mycotic aneurysms when combined with magnetic resonance angiography. Cerebral microbleeds are small, round hypointense lesions best visualized on T2*-weighted sequences. They are present in up to 50% to 60% of patients with IE, more frequently than in the general population.[52,53] Microbleeds are usually asymptomatic, and their presence reflects small vessel injury or fragility. Cerebral microbleeds are not considered a minor diagnostic criterion.

Brain MRI is needed in patients with neurologic symptoms. In those who are neurologically asymptomatic and require cardiac surgery, obtaining a preoperative brain MRI is reasonable. Detection of silent ischemic emboli is not a reason to delay cardiac surgery as these findings have not been associated with worse outcomes. Obtaining a brain MRI in asymptomatic patients not requiring cardiac surgery is of unclear benefit and discussion with the endocarditis team is warranted.

Key Take-Home Points

- Brain MRI is superior to CT in patients who have IE and neurologic symptoms.
- Brain imaging, preferably MRI, is reasonable to obtain in neurologically asymptomatic patients before cardiac surgery.
- The endocarditis team needs to be involved in the decision to obtain brain MRI in neurologically asymptomatic patients.

CT Chest/Abdomen/Pelvis Imaging

Aside from directly diagnosing cardiac lesions, CT chest/abdomen/pelvis can also be used to detect 2023 Duke/ISCVID minor criteria, which can assist in diagnosis, including septic emboli (eg, lungs, spleen, kidneys); abscesses (eg, splenic, renal, cerebral); mycotic aneurysms (CT angiogram); infarctions, particularly splenic; and spondylodiscitis and osteoarticular infections.

CT can also be used to assess the aorta and femoral vessels for surgical access, and can provide information on the morphology, size, and calcification of the aortic valve and mitral valve annulus, aortic root, and ascending aorta, which can affect surgical feasibility. The American Association for Thoracic Surgery states that routine preoperative brain CT, even in the absence of symptoms, is justified as part of the workup.[54]

A multicenter evaluation of routine abdominal CT in patients with left-sided IE found that about one-third of patients had embolic lesions involving the spleen, liver, or kidneys; however, the presence of these lesions was not associated with changes in clinical management or outcomes.[55]

Key Take-Home Points

- CT chest/abdomen/pelvis may demonstrate extracardiac findings that support an IE diagnosis.
- Routine CT chest/abdomen/pelvis in asymptomatic patients is unlikely to be of benefit and its use should be reserved for symptomatic patients.

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TREATMENT

Intravenous antimicrobial therapy remains the cornerstone of treatment during the initial phase of IE. During the continuation phase, antimicrobial treatment options may include continued intravenous therapy, POT, and long-acting injectable lipoglycopeptides. Antimicrobial regimens are provided for the most common causes of IE (eg, staphylococci, streptococci, enterococci) along with select Gram-negative and fungal organisms (Tables 1–4).[2,56–62]

Candidates for POT

The POET trial by Iversen et al[2] included 400 patients with left-sided IE due to staphylococci (excluding methicillin-resistant *S aureus*), streptococci, or *E faecalis* initially treated with intravenous antibiotics then randomized to continue intravenous treatment or transition to oral antibiotics. They showed that the primary 6-month composite outcome (all-cause death, unplanned cardiac surgery, embolic events, or relapse of bacteremia) occurred less commonly in the oral treatment group compared with the intravenous treatment group, meeting noninferiority. The criteria required for transition to POT are listed in Table 5.[2]

The POET results challenge the longstanding notion that prolonged intravenous antibiotics delivery is the only route of IE treatment. A 2025 survey of clinicians in the United States found that adoption of POT for IE is used sparingly, with use varying according to the specific pathogen and availability of an active oral agent. Barriers to POT included concerns about relapse, adherence concerns, and insufficient evidence, with 75% of respondents expressing a desire for clearer guidelines and additional data.[63] Similar results were seen in an international survey.[64] The 2023 ESC guidelines for the management of IE include oral antibiotic therapy as an option after the early critical phase of treatment, starting on day 10 of intravenous therapy or 10 days after cardiac surgery, for patients with negative blood cultures and TEE ruling out severe valvular regurgitation and prosthetic valve dysfunction, with any causative agent except highly difficult-to-treat microorganisms.[16]

The writing group recognizes that POT may be an appropriate management strategy for select patients who qualify and meet all stabilization criteria. During the initial treatment phase, intravenous antibiotics are administered until the patient has achieved clinical stability, including resolution of a sepsis syndrome, sterile blood cultures, TEE without concerning findings requiring acute surgical intervention, available antimicrobial susceptibilities, and the ability to absorb and tolerate oral antibiotic therapies. Recruitment of patients into POET was prolonged, and 80% of patients were denied enrollment as they did not fulfill inclusion criteria.[2] In addition, only 26% of patients in both trial arms had PVE, and a total of 5 had PWID. These factors have been considered limitations and have prompted questions regarding the applicability of POT in IE.

Treatment with 2 active agents that were used in the POET trial was favored by the majority of writing group members. However, some writing group members considered using monotherapy in select scenarios. In a recently published study of POT (ENDO-ORAL), a propensity score analysis used to determine treatment failure in a retrospective cohort found no significant difference between groups (exclusively intravenous treatment or a switch to oral therapy).[65] In 26% of patients, oral amoxicillin monotherapy was used.

When a peripherally inserted central catheter is not feasible (eg, housing challenges, prohibitive costs, patient preference), it is reasonable to consider POT after an initial inpatient intravenous phase. Use of POT is not considered inferior or second-line therapy, but rather an option for select patients after the acute phase of IE treatment. Given the inherent complexities in IE management, the involvement of an infectious diseases specialist is essential to determine potential treatment regimens.

Key Take-Home Points

- POT is an option for select patients after the acute phase of IE treatment.
- Consensus was not reached within the writing group on dual versus monotherapy.
- Transition to POT is based on multiple factors that are considered by an endocarditis team.

Role for Long-Acting Injectable Lipoglycopeptides

The role of long-acting injectable lipoglycopeptides (dalbavancin and oritavancin) for IE treatment has evolved over the past decade. Their use is favored given the long half-life (>200 hours), making it desirable for patients in whom daily intravenous therapy may be difficult or prohibitive, or in whom POT is not an option. In a systematic review of dalbavancin as transitional therapy for IE, the majority of studies included a clinical failure rate <10% and had a favorable safety profile.[66] In one single-center retrospective cohort, 61 older patients with IE and clinically significant comorbidities who received dalbavancin for IE due to Gram-positive cocci had a total reduction of hospitalization of 1090 days, and a clinical cure rate of 86.9%, with a small number of adverse drug events.[67]

A major drawback of these agents is their high costs, although they may ultimately be cost-effective by reducing hospital length of stay.[68] The recent cost reduction of dalbavancin may lower this barrier. However, its use

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Table 1. Comparison of the 2015 and 2026 American Heart Association Antimicrobial Regimens for Infective Endocarditis Due to Staphylococci

Organism and valve type	2015 AHA guideline regimen	2026 AHA-recommended initial therapy*	2026 AHA potential alternative continuation regimens	
			Long-acting lipoglycopeptides based on susceptibility testing (start after clearance of bacteremia during continuation phase)	Oral continuation regimens according to POET, based on susceptibility testing (start after >10 d IV therapy during continuation phase)†
Methicillin-susceptible, native valve	Nafcillin/oxacillin 12 g/24 h IV in 6 equally divided doses For penicillin-allergic patients (non–anaphylactoid type): cefazolin 6 g/24 h IV in 3 equally divided doses	Nafcillin/oxacillin 12 g/24 h IV in 6 equally divided doses OR Cefazolin 6 g/24 h IV in 3 equally divided doses	Dalbavancin 1500 mg IV on d 1 and 1500 mg on d 8[56] OR Alternate dalbavancin dosing regimen: 1500 mg on d 1 and d 15‡ OR Oritavancin 1200 mg IV on d 1 and 1200 mg on d 8	Dicloxacillin 1 g QID plus rifampin 600 mg BID OR Dicloxacillin 1 g QID plus fusidic acid 750 mg BID§ OR Linezolid 600 mg BID plus rifampin 600 mg BID‖ OR Linezolid 600 mg BID plus fusidic acid 750 mg BID
Methicillin-susceptible, prosthetic valve	Nafcillin/oxacillin 12 g/24 h IV in 6 equally divided doses PLUS Rifampin 900 mg per 24 h IV or orally in 3 equally divided doses PLUS Gentamicin 3 mg/kg per 24 h IV or IM in 2 or 3 equally divided doses (2 wk)	Nafcillin/oxacillin 12 g/24 h IV in 6 equally divided doses OR Cefazolin 6 g/24 h IV in 3 equally divided doses PLUS Rifampin 900 mg per 24 h IV or orally in 3 equally divided doses¶	No data	POET did not distinguish between native vs prosthetic valve regarding regimens
Methicillin-resistant, native valve	Vancomycin 30 mg/kg per 24 h IV in 2 equally divided doses OR Daptomycin ≥8 mg·kg·dose	Vancomycin hydrochloride 30 mg/kg per 24 h IV in 2 equally divided doses OR Daptomycin 8–12 mg/kg IV daily# OR Ceftaroline 600 mg IV q 8 h OR Linezolid 600 mg IV or orally every 12 h	Dalbavancin 1500 mg IV on d 1 and 1500 mg on d 8 Alternate dalbavancin dosing regimen: 1500 mg IV on d 1 and d 15	Linezolid 600 mg BID plus rifampin 600 mg BID‖ OR Linezolid 600 mg BID plus fusidic acid 750 mg BID§
Methicillin-resistant, prosthetic valve	Vancomycin 30 mg/kg per 24 h IV in 2 equally divided doses PLUS Rifampin 900 mg per 24 h IV or orally in 3 equally divided doses PLUS Gentamicin 3 mg/kg per 24 h IV or IM in 2 or 3 equally divided doses (2 wk)	Vancomycin 30 mg/kg per 24 h IV in 2 equally divided doses OR Daptomycin 8–12 mg/kg IV daily# OR Ceftaroline 600 mg IV q 8 h OR Linezolid 600 mg IV or orally every 12 h PLUS Rifampin 900 mg per 24 h IV or orally in 3 equally divided doses¶	No data	POET did not distinguish between native vs prosthetic valves regarding regimens
Duration of therapy	6 wk for all cases 2 wk for gentamicin synergy	Native valve 4 wk; prosthetic valve 6 wk Duration reflects the cumulative pathogen-directed antimicrobial therapy provided after clearance of bacteremia; if long-acting lipoglycopeptides are used, a single dose would be estimated to provide 2 wk of antimicrobial duration and a 2-dose series to provide 6 wk of effective therapy		

Derived from Iversen et al,[2] Turner et al,[56] Pries-Heje et al,[57] Freling et al,[58] Senneville et al,[59] and Cojutti et al.[60] AHA indicates American Heart Association.

*Intravenous antimicrobial therapy regimens can be used for the entire duration of treatment or limited to the initial phase of therapy followed by a long-acting lipoglycopeptide or oral regimen. Infection severity, source control, and patient adherence should be considered in choosing a continuation regimen. Some writing group members would consider monotherapy given patient-specific circumstances. Oral therapy should be limited to the continuation phase of treatment in those who fulfill stabilization criteria as shown in Table 5, and an individualized approach to oral therapy determined by the endocarditis team.

†Combination therapy was evaluated in the POET trial (Partial Oral Treatment of Endocarditis),[2] although no methicillin-resistant *Staphylococcus aureus* infective endocarditis cases were included in the randomized cohort. Fixed-dose oral antibiotics may risk underdosing for individuals with morbid obesity; consider dose modification based on therapeutic drug monitoring. Other published regimens used in retrospective cohorts and prospective registries have included either monotherapy or combination therapy regimens. Input from the endocarditis team is advised in methicillin-resistant *S aureus* infective endocarditis cases.[57,58]

‡Therapeutic drug monitoring of dalbavancin may be considered, if available, for longer treatment durations or in cases of renal failure.[59]

§Fusidic acid is unavailable in the United States.

‖When linezolid is used in combination with rifampin, this can result in lower bloodstream concentrations of linezolid. Therapeutic drug monitoring can be considered for linezolid, with trough concentration goals of 2 to 8 mg/L.[60]

¶Delay the initiation of rifampin until bacteremia has cleared and after 5 to 7 days of antibiotics. Use of rifampin is to be considered on an individual basis given toxicity profile and drug–drug interactions.

#Some writing group members use daptomycin in combination with β-lactams or IV fosfomycin at least until blood cultures have cleared for purposes of synergy and reducing likelihood of antimicrobial resistance.

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Table 2. Comparison of the 2015 and 2026 American Heart Association Antimicrobial Regimens for Infective Endocarditis Due to Viridans Group Streptococci and *Streptococcus gallolyticus*

Organism and valve type	2015 AHA guidelines for reference	2026 AHA initial therapy*	2026 AHA potential alternative continuation regimens	
			Long-acting lipoglycopeptides based on susceptibility testing (start after clearance of bacteremia during continuation phase)	Oral continuation regimens according to POET, based on susceptibility testing (start after ≥10 d IV therapy during continuation phase)†
Penicillin-susceptible VGS/*S gallolyticus* (MIC <0.12 mg/L)	Aqueous crystalline penicillin G sodium 24 million U/24 h IV either continuously or in 4–6 equally divided doses OR Ceftriaxone 2 g/24 h IV or IM in 1 dose WITH OR WITHOUT Gentamicin 3 mg·kg·d IV or IM given in 1 dose (2 wk)	Aqueous crystalline penicillin G sodium 24 million U/24 h IV either continuously or in 4–6 equally divided doses OR Ceftriaxone 2 g/24 h IV or in 1 dose	Dalbavancin 1500 mg IV on d 1 and 1500 mg on d 8‡ OR Alternate dalbavancin dosing regimen: 1500 mg on d 1 and d 15‡ OR Oritavancin 1200 mg IV on d 1 and 1200 mg on d 8	Amoxicillin 1 g QID plus rifampin 600 mg BID OR Linezolid 600 mg BID plus rifampin 600 mg BID§ OR Linezolid 600 mg BID plus moxifloxacin 400 mg QD
Unable to tolerate β-lactams	Vancomycin 30 mg/kg per 24 h IV in 2 equally divided doses	Vancomycin hydrochloride 30 mg/kg per 24 h IV in 2 equally divided doses OR Linezolid 600 IV every 12 h		Linezolid 600 mg BID plus rifampin 600 mg BID§ OR Moxifloxacin 400 mg QD plus rifampin 600 mg BID OR Moxifloxacin 400 mg QD plus clindamycin 600 mg TID
Penicillin-intermediate strain (MIC >0.25–2 mg/L and ceftriaxone MIC >0.5 mg/L)	Aqueous crystalline penicillin G sodium 24 million U/24 h IV either continuously or in 4–6 equally divided doses OR Ceftriaxone 2 g/24 h IV or IM in 1 dose PLUS Gentamicin 3 mg·kg·d IV or IM given in 1 dose	Ceftriaxone 2 g/24 h IV or IM in 1 dose		
Penicillin-resistant strain (MIC ≥4.0 mg/L)	Aqueous crystalline penicillin G sodium 24 million U/24 h IV either continuously or in 4–6 equally divided doses OR Ceftriaxone 2 g/24 h IV or IM in 1 dose PLUS Gentamicin 3 mg·kg·d IV or IM given in 1 dose	Ceftriaxone 2 g/24 h IV or IM in 1 dose PLUS Gentamicin 3 mg·kg·d IV or IM daily for the first 2 wk OR Vancomycin hydrochloride 30 mg/kg per 24 h IV in 2 equally divided doses		
Duration of therapy	4 wk with native valve ≥6 wk with prosthetic valve or prosthetic material	Native valve 4 wk; prosthetic valve 6 wk Duration reflects the cumulative pathogen-directed antimicrobial therapy provided after clearance of bacteremia; if long-acting lipoglycopeptides are used, a single dose would be estimated to provide 2 wk of antimicrobial duration and a 2-dose series to provide 6 wk of effective therapy		

Derived from Iversen et al,[2] Pries-Heje et al,[57] Freling et al,[58] Senneville et al,[59] and Cojutti et al.[60] AHA indicates American Heart Association; MIC, minimum inhibitory concentration; and VGS, viridans group streptococci.

*Intravenous antimicrobial therapy regimens can be used for the entire duration of treatment or limited to the acute initial phase of therapy followed by a long-acting lipoglycopeptide or oral regimen. Oral therapy should be limited to the continuation phase of treatment in those who fulfill stabilization criteria as shown in Table 5, and an individualized approach to oral therapy determined by the endocarditis team.

†Combination oral therapy was studied in the POET trial (Partial Oral Treatment of Endocarditis)[2]; no clear data exist on whether combination vs monotherapy is required for oral continuation regimens. Some writing group members would consider monotherapy given patient-specific circumstances. Fixed-dose oral antibiotics may risk underdosing for individuals with morbid obesity; consider dose modification based on therapeutic drug monitoring. Potential regimens are listed in alphabetical order given lack of data on superiority of any specific regimen or combination.[57,58]

‡Therapeutic drug monitoring of dalbavancin, if available, may be considered for longer treatment durations or in cases of renal failure.[59]

§When linezolid is used in combination with rifampin, this can result in lower bloodstream concentrations of linezolid. Therapeutic drug monitoring can be considered for linezolid, with trough concentration goals of 2 to 8 mg/L.[60]

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Table 3. Comparison of the 2015 and 2026 American Heart Association Antimicrobial Regimens for Infective Endocarditis Due to *Enterococcus faecalis* or *Enterococcus faecium*

Organism and valve type	2015 AHA guidelines for reference	2026 AHA initial therapy*	2026 AHA potential alternative continuation regimens	
			Long-acting lipoglycopeptides based on susceptibility testing (start after clearance of bacteremia during continuation phase)	Oral continuation regimens according to POET, based on susceptibility testing (start after >10 d IV therapy during continuation phase)†
E faecalis	Ampicillin 2 g q 4 h IV OR 18–30 million U/24 h IV either continuously or in 6 equally divided doses PLUS Gentamicin 3 mg·kg·d IV or IM given in 2–3 equally divided doses	Ampicillin 2 g IV q 4 h PLUS Ceftriaxone 2 g IV BID	Dalbavancin 1500 mg IV on d 1 and 1500 mg on d 8‡ OR Alternate dalbavancin dosing regimen: 1500 mg on d 1 and d 15‡ OR Oritavancin 1200 mg IV on d 1 and 1200 mg on d 8	Amoxicillin 1 g QID plus rifampin 600 mg BID OR Amoxicillin 1 g QID plus moxifloxacin 400 mg QD OR Linezolid 600 mg BID plus rifampin 600 mg BID§ OR Linezolid 600 mg BID plus moxifloxacin 400 mg QD
Unable to tolerate β-lactams, penicillin resistance; intrinsic or β-lactamase producer	Vancomycin 30 mg/kg per 24 h IV in 2 equally divided doses PLUS Gentamicin 3 mg·kg·d IV or IM given in 3 divided doses	Vancomycin 30 mg/kg per 24 h IV in 2 equally divided doses OR Linezolid 600 mg IV/PO BID OR Daptomycin 10–12 mg/kg QD		Linezolid 600 mg BID plus moxifloxacin 400 mg QD OR Linezolid 600 mg BID plus rifampin 600 mg BID§
E faecium (vancomycin-resistant)	Linezolid IV or orally every 12 h OR Daptomycin 10–12 mg/kg per dose	Linezolid 600 mg IV/PO BID OR Daptomycin 10–12 mg/kg QD in combination with a β-lactam‖	Not advised due to nonsusceptibility in vancomycin-resistant strains	Not included in POET
Duration of therapy	4 wk for patients with symptoms of illness <3 mo ≥6 wk if >3 mo and with prosthetic valve or prosthetic material	Native valve 6 wk; prosthetic valve 6 wk Duration reflects the cumulative pathogen-directed antimicrobial therapy provided after clearance of bacteremia; if long-acting lipoglycopeptides are used, a single dose would be estimated to provide 2 wk of antimicrobial duration and a 2-dose series to provide 6 wk of effective therapy		

Derived from Iversen et al,[2] Pries-Heje et al,[57] Freling et al,[58] Senneville et al,[59] Cojutti et al,[60] and Chuang et al,[61] AHA indicates American Heart Association.

*Intravenous antimicrobial therapy regimens can be used to complete the entire duration of treatment or limited to the initial phase of therapy followed by a long-acting lipoglycopeptide or oral regimen. Oral therapy would be limited to the continuation phase of treatment in those who fulfill stabilization criteria as shown in Table 5, and an individualized approach to oral therapy determined by the endocarditis team.

†Combination oral therapy was studied in the POET trial (Partial Oral Treatment of Endocarditis)[2]; no clear data exist on whether combination vs monotherapy is required for oral continuation regimens. Some writing group members would consider monotherapy given patient-specific circumstances. Potential regimens are listed in alphabetical order given lack of data on superiority of any specific regimen or combination.[57,58]

‡Therapeutic drug monitoring of dalbavancin, if available, may be considered for longer treatment durations or in cases of renal failure.[59]

§When linezolid is used in combination with rifampin, this can result in lower bloodstream concentrations of linezolid. Therapeutic drug monitoring can be considered for linezolid, with trough concentration goals of 2 to 8 mg/L.[60]

‖Choice of combination β-lactam based on patient-specific factors.[61] Common combination agents include ampicillin, ceftaroline, or ertapenem. Can consider IV fosfomycin if β-lactam is contraindicated.

remains off-label for IE treatment. In addition, there have been case reports of the emergence of dalbavancin-, vancomycin-, and daptomycin-nonsusceptible *S aureus* and lipoglycopeptide cross-resistance arising in patients treated with dalbavancin.[69–71] The rate of relapse in *E faecalis* IE is high at 8% (and 15% in PVE).[72]

Suggested regimens are included in Tables 1 through 3. Common IE regimens of dalbavancin that provide cover for 4 to 6 weeks include a 2-dose course with 1500 mg on day 1 and day 8 as supported by the DOTS trial (Dalbavancin as an Option for Treatment of *Staphylococcus aureus* Bacteremia) for complicated *S aureus* bacteremia, which included individuals with right-sided IE,[56] or a total 3000 mg over a 4-week period with 1500 mg administered at day 0 and day 15.[59]

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Table 4. Comparison of 2015 and 2026 American Heart Association Antimicrobial Regimens for Infective Endocarditis Due to Select Gram-Negative Bacteria and Fungi

Organism and valve type	2015 AHA guidelines for reference	2026 AHA initial therapy	2026 AHA oral continuation regimens based on susceptibility testing (start after source control attained or clinical stability)
HACEK	Ceftriaxone 2 g IV q 24 h OR Ampicillin 2 g IV every 4 h OR Ciprofloxacin 1000 mg/24 h orally or 800 mg/24 IV in 2 equally divided doses	Ceftriaxone 2 g IV QD* OR Ampicillin 2 g IV q 4 h OR Ciprofloxacin 400 mg IV BID	Levofloxacin 750 mg QD†
Pseudomonas aeruginosa	Antipseudomonal β-lactam PLUS Aminoglycoside OR Fluoroquinolone	Cefepime 2 g IV TID OR Piperacillin-tazobactam 4.5 g QID OR Meropenem 1 g TID PLUS Either tobramycin 8 mg/kg IV QD (goal peak and trough concentrations of 15 to 20 µg/mL and ≤2 µg/mL) or a fluoroquinolone§	Levofloxacin 750 mg QD‡
Candida	Amphotericin B (lipid formulation) PLUS Flucytosine	Amphotericin B (lipid formulation) 3–5 mg/kg IV QD PLUS Flucytosine 25 mg/kg PO q 6 h OR Micafungin 150 mg IV q 24 h	Oral azole therapy based on susceptibility testing‖
Aspergillus	Amphotericin B (lipid formulation)	Voriconazole loading dose: 6 mg/kg IV BID for 2 doses; maintenance dose 4 mg/kg IV BID¶ OR Amphotericin B (lipid formulation) 3–5 mg/kg IV QD	Oral azole with activity against *Aspergillus* species
Duration of therapy, wk	4 for HACEK NVE 6 for HACEK PVE 6 minimum for pseudomonal NVE and PVE 6 for fungal endocarditis NVE and PVE	HACEK native valve: 4# HACEK prosthetic valve: 6 Pseudomonal native valve: 6 Pseudomonal prosthetic valve: 6 Fungal endocarditis native valve: 6 Fungal endocarditis prosthetic valve: 6	

Derived from Schuetz et al.[62] AHA indicates American Heart Association; NVE, native valve endocarditis; and PVE, prosthetic valve endocarditis.

*Ceftriaxone is preferred over other listed options if organism is susceptible.

†No randomized controlled trial data. Oral fluoroquinolone step-down possible in select stable patients if organism susceptible. Levofloxacin or moxifloxacin may be substituted.

‡No data available on optimal duration of combination therapy. Combination therapy can be considered until susceptibilities are available.

§Some writing group members would maintain combination therapy that includes intravenous antibiotics plus oral fluoroquinolone. Changes to the aminoglycoside (gentamicin, tobramycin, and amikacin) breakpoints were published in the Clinical and Laboratory Standards Institute Performance Standards for Antimicrobial Susceptibility Testing: M100, 32nd and 33rd editions,[62] eliminating gentamicin as a suggested treatment option for *P aeruginosa*, as the data demonstrated that wild-type isolates (with minimum inhibitory concentration below the epidemiologic cutoff value) were not treatable with gentamicin, which raises the possibility of intrinsic resistance.

‖Transition to oral therapy can be considered after clearance of candidemia, clinical improvement, and surgical source control. Combined medical and surgical management is key to improving outcomes. In patients who are not surgical candidates, transition to long-term suppressive therapy is a frequent approach once initial treatment has been completed.

¶Therapeutic drug monitoring of voriconazole and posaconazole, if available.

#No data are available on outcomes comparing 4 vs 6 weeks of therapy for native-valve HACEK (*Haemophilus*, *Aggregatibacter*, *Cardiobacterium*, *Eikenella*, and *Kingella*) infective endocarditis; treatment duration should be guided by clinical judgment and infection severity.

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Table 5. Criteria for Partial Oral Therapy and Long-Acting Injectable Antimicrobials

Type of antimicrobial home therapy	Partial oral therapy	Long-acting injectable antimicrobials
Inclusion criteria	IE caused by streptococci, *E faecalis*, methicillin-susceptible *Staphylococcus aureus*, or coagulase-negative staphylococci; receipt of appropriate IV antimicrobials for ≥10 d and ≥7 d after valve surgery; and clinically stable	IE caused by streptococci, enterococci, or staphylococci
	Antimicrobial response with no fever for ≥3 d, CRP <25% of peak value or <20 mg/L, and leukocytes <15×109/L	Initial clinical response to IV antimicrobials (see partial oral therapy criteria)
	TEE performed within 2 d of the transition to oral antimicrobials, without evidence of infection progression, abscess formation, or a new indication for cardiac surgery, and should document vegetation size and characteristics	Unable to receive traditional OPAT due to lack of IV access, housing instability, or active or recent injection drug use
Exclusion criteria	Body mass index >40	Cost
	Suspicion of reduced absorption of oral treatment due to gastrointestinal disorder	Lack of availability
	Concomitant infection requiring IV antimicrobial therapy (OPAT)	Drug–drug interactions
	Reduced compliance	Drug allergy
Comments	Limited number of PVE cases included in POET	Therapeutic drug monitoring is anticipated for future use

CRP indicates C-reactive protein; IE, infective endocarditis; OPAT, outpatient parenteral antibiotic therapy; and POET, Partial Oral Treatment of Endocarditis.[2]

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The most common regimen for oritavancin is 1200 mg weekly for 2 to 6 doses. In a systematic review on the clinical efficacy of multidose oritavancin, which included 16 studies and 301 participants treated with oritavancin for osteomyelitis and skin and soft-tissue infections, a cure rate of 85% was reported, and oritavancin was well-tolerated, with a favorable safety profile.[73] Oritavancin interferes with several coagulation assays, causing artificial prolongation of activated partial thromboplastin time for up to 120 hours, prothrombin time/international normalized ratio for up to 12 hours, and activated clotting time for up to 24 hours after administration. Therefore, intravenous unfractionated heparin is contraindicated for 120 hours after oritavancin dosing.[74] Oritavancin retains activity against vancomycin-resistant enterococci that have the *VanA* resistance gene, whereas dalbavancin lacks activity against isolates harboring *VanA*.[75] Outpatient parenteral antibiotic therapy (OPAT) criteria for transitioning to long-acting injectable lipoglycopeptides are listed in Table 5.[2]

Key Take-Home Points

- Long-acting injectable lipoglycopeptides (eg, dalbavancin, oritavancin) represent an alternative antimicrobial treatment option in select patients.
- Use of long-acting injectable lipoglycopeptides may be associated with increased relapse rates, particularly in nonoperated prosthetic valve *E faecalis* IE.
- Emergence of cross-resistance of dalbavancin-, vancomycin-, and daptomycin-nonsusceptible *S aureus* with lipoglycopeptides is of concern.

Role of OPAT in IE

Traditional OPAT has many benefits, as detailed in the 2015 AHA IE scientific statement.[3] Recently, a GAMES study (Grupos de Apoyo al Manejo de la Endocarditis en España) expanded the OPAT IE criteria, which thereafter were validated and incorporated into the 2023 ESC IE guidelines.[16,76,77] Before OPAT is considered, several conditions are needed to provide a safe and successful program. Endocarditis team members are needed in the outpatient setting to permit ongoing evaluation, antimicrobial administration (eg, home, infusion therapy centers, skilled nursing facilities), and laboratory monitoring.

CONTROVERSIAL TOPICS

Gentamicin and Rifampin in the Management of Staphylococcal PVE

Adjunctive gentamicin and rifampin historically has been recommended for the treatment of PVE caused by *S aureus* and coagulase-negative staphylococci.[3,16] This was based on in vitro studies, results of animal models of experimental IE, and retrospective cohort studies of *S epidermidis* PVE from the 1980s.[78,79] Adjunctive gentamicin was thought to confer a clinical benefit, as it exhibits synergistic, bactericidal activity when administered with either β-lactams or vancomycin and, unlike β-lactams and glycopeptides,[80] diffuses homogeneously into the valvular vegetation,[81] thereby presumably preventing development of rifampin resistance. The addition of rifampin was thought to be key, based on its greater intrinsic activity in Gram-positive biofilm-associated infections,[82] such as staphylococcal PVE. Kitano and Yoshida[83] conducted a retrospective analysis of a large database of patients with *S aureus* PVE treated with a rifampicin-containing regimen without gentamicin compared with a gentamicin-containing regimen without rifampicin, and found that the rifampicin-containing regimen without gentamicin was associated with lower 1-year mortality rates.

Evidence from more recent clinical studies casts doubt on the clinical utility of triple antibiotic therapy.[83–86]

Among 334 patients with staphylococcal PVE, the GAMES study found no difference in mortality rates between those treated with or without adjunctive gentamicin, although findings were limited by lack of adjustment for indication bias and by the relatively small number of patients (n=17) who did not receive adjunctive aminoglycoside.[84] A similar analysis of adjunctive rifampin among 180 patients with staphylococcal PVE also found no apparent harm in the 79 patients who did not receive rifampin.[85] This study was also limited by its retrospective design and the lack of indication bias adjustment. A US Veterans Health Administration study corroborated the conclusions of the 2 previous reports. More than half of the 373 patients with staphylococcal PVE (56%) either died or relapsed within 1 year after diagnosis, and neither adjunctive rifampin or gentamicin was associated with improved outcomes.[86]

The lack of benefit was further supported by a recent meta-analysis of 343 patients with IE (44.9% [n = 154] of whom had *S aureus*) from 4 eligible studies.[87] Whereas neither the addition of gentamicin (odds ratio, 0.98 [95% CI, 0.39–2.46]) or rifampin (odds ratio, 1.29 [95% CI, 0.71–2.33]) to combination therapy reduced clinical failure, gentamicin with rifampin was associated with nephrotoxicity, hepatotoxicity, and drug–drug interactions necessitating discontinuation of rifampin in approximately one-third of patients.[87]

Key Take-Home Points

- Consensus was reached that the risk of gentamicin outweighs any potential benefits in staphylococcal PVE.
- Consensus was not reached regarding the use of rifampin; discussion with the endocarditis team is warranted.

Addition of Gentamicin in Cases of Viridans Group Streptococci or *Streptococcus gallolyticus* IE With Intermediate Resistance to Penicillin

There are no RCT data to inform the benefit of adding gentamicin in this setting. Both the 2015 AHA scientific statement[3] and the 2023 ESC guidelines[16] for IE recommend treating IE caused by viridans group streptococci (VGS) or *S gallolyticus* with intermediate resistance to penicillin (minimum inhibitory concentration [MIC] 0.25–2 mg/L) with a β-lactam antibiotic (penicillin, ampicillin, or ceftriaxone) for 4 to 6 weeks, with gentamicin added for the first 2 weeks.

A recent study by the GAMES group analyzed 914 consecutive cases of definite IE caused by VGS or *S gallolyticus*, 226 of which (25%) had intermediate susceptibility to penicillin (intermediate resistance to penicillin MIC 0.25–2 mg/dL).[88] Monotherapy was used in 415 cases (45%; cephalosporin in 331 cases), and 499 cases (55%) received combination therapy. Monotherapy with β-lactam antibiotics (80% received ceftriaxone) was not associated with an increase of in-hospital mortality in patients with intermediate resistance to penicillin (odds ratio, 1.01 [95% CI, 0.26–3.96]; *P*=0.982), although these patients had a higher risk of death at 1 year of follow-up (2.96 [1.12–7.83]; *P*=0.028).[88]

A 2025 Société de Pathologie Infectieuse de Langue Française–Association pour l'Étude et la Prévention de l'Endocardite Infectieuse position statement endorsed the use of β-lactam therapy based on the MIC in both VGS and *S gallolyticus* IE.[89] For patients with intermediate resistance to penicillin, the final treatment is based on the MIC of 3rd-generation cephalosporins. If the 3rd-generation cephalosporin MIC is ≤0.5 mg/L (susceptible), intravenous ceftriaxone without gentamicin can be used. Those with a 3rd-generation cephalosporin MIC >0.5 mg/L (intermediate or resistant) would receive amoxicillin/ampicillin for 4 to 6 weeks plus gentamicin the first 2 weeks.[89]

Key Take-Home Points

- Consensus was reached regarding the use of ceftriaxone monotherapy for VGS or *S gallolyticus* IE with intermediate resistance to penicillin.
- Consensus was reached regarding the addition of gentamicin for the first 2 weeks to ceftriaxone in patients with fully penicillin-resistant isolates of VGS and *S gallolyticus*.

Combination Therapy for *E faecalis* IE

Penicillin-susceptible *E faecalis* IE is treated with a combination of ampicillin (or amoxicillin) plus gentamicin or ampicillin (or amoxicillin) plus ceftriaxone, the latter of which is favored by the writing group. Although clinical evidence supporting the need for combination therapy versus monotherapy is limited, available evidence indicates that the relapse rate after treatment with monotherapy is high.[90] For example, a multicenter retrospective study across 14 hospitals in France evaluated 279 patients treated with combination versus monotherapy. The monotherapy group had a high rate of relapse (46.2%) and lower rates with different combinations (4.3%–14.7%), supporting the use of combination therapy in *E faecalis* IE.[91]

Key Take-Home Points

- Combination therapy is associated with lower rates of relapse in *E faecalis* IE compared with monotherapy.
- Consensus was reached supporting the use of combination therapy in *E faecalis* IE.

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The Association of Colon Neoplasia and *E faecalis* IE

Although the association of bacteremia caused by *S gallolyticus* (*S bovis* biotype I) and colorectal cancer is well known,[92] it is increasingly evident that *E faecalis* is also associated with benign or malignant colon pathology in most patients with IE, regardless of whether the focus of the infection is known.[93–95] Therefore, in patients with these pathogeneses, a colonoscopy would be reasonable to perform.

Key Take-Home Point

- Colonoscopy is reasonable in patients diagnosed with *S gallolyticus* or *E faecalis* IE to evaluate for colonic malignancy. (Consensus was reached.)

Duration of Antibiotic Therapy After Surgical Valve Replacement

Despite previous opinions that antimicrobial therapy must be completed before cardiac valve surgery, early (48–72 hours) cardiac valve surgery has been shown to reduce embolic events in patients with complicated left-sided IE and improve outcomes.[96] The results of this RCT (EASE [Early Surgery Versus Conventional Treatment in Infective Endocarditis]) have influenced cardiac valve surgery timing relative to antimicrobial administration. As a result of this change, the postoperative duration of antimicrobial therapy has been extended.

Patients with IE generally receive 4 to 6 weeks of antimicrobial therapy despite the lack of RCT data to define the optimal duration of therapy. In patients who undergo cardiac valve surgery, the duration of postoperative antimicrobial therapy may be influenced by various clinical factors. These include the timing of surgery relative to the initiation and duration of preoperative antimicrobial therapy, intraoperative findings of perivalvular abscess, positive cultures from resected valve tissue, positive Gram stain on histopathology in the absence of positive valve cultures, and positive molecular diagnostic test results.

The 2015 AHA scientific statement[3] determined that it was reasonable to provide an entire course of antimicrobial therapy (4–6 weeks) if resected valve tissue culture was positive or if a perivalvular abscess was found. If valve tissue cultures were negative, regardless of a positive Gram stain on histopathology, then the recommendation was to complete the planned duration for native or prosthetic valve infection, less the number of days of antimicrobial therapy received preoperatively.[3]

Recent studies have challenged this approach. Johansson et al[97] found no association between positive valve cultures and 1-year mortality or relapse. Kim et al[98] compared recurrence and mortality rates of 216 patients who underwent surgical valve replacement who received >2 weeks or <2 weeks of antimicrobial therapy postoperatively, and found no difference in relapse rate, 1-year recurrence, or 1-year mortality between the 2 groups. Rao et al[99] evaluated 182 patients with IE who underwent valve surgery, with a median duration of postoperative antimicrobial therapy of 23.5 days, which decreased significantly over the study period, and noted low rates of relapse (1.1%) and postoperative deaths at 1 year (9.9%); the duration of postoperative antimicrobial therapy did not affect the rate of relapse of IE or survival at 1 year.

These findings suggest that surgical management of IE provides considerable source control and antimicrobial therapy beyond 2 weeks after surgical valve replacement may be of limited added benefit. It would be reasonable to consider 2 weeks of antimicrobial therapy after surgical valve replacement regardless of valve tissue culture results.

It is also important to recognize the timing of surgery in relation to the amount of preoperative therapy received. Patients who undergo valve surgery before completing antimicrobial therapy should receive the planned 4- to 6-week course, with preoperative treatment days counted toward the total duration, provided that at least 14 days of therapy are administered postoperatively, avoiding "restarting the clock" after surgery with an additional 4 to 6 weeks of therapy. If the patient completes therapy before cardiac valve surgery and there are no signs of active infection, including negative valve tissue cultures, then no additional antibiotics are needed postoperatively. In select patients, a longer course of antimicrobial therapy may be determined by the endocarditis team. Metastatic foci of infection may dictate prolonging therapy.

When resected valve tissue is screened by molecular diagnostics, the detection of DNA does not necessarily indicate the persistence of viable organisms[100]; therefore, these findings would not affect postoperative duration of antimicrobial therapy.

Key Take-Home Points

- Completion of antimicrobial therapy before cardiac valve surgery is not required.
- Resected valve tissue culture or molecular diagnostic positivity does not influence the duration of antimicrobial therapy postoperatively.
- Patients who undergo cardiac valve surgery before completing antimicrobial therapy for IE should receive the planned 4- to 6-week course, less the number of days received preoperatively, provided that at least 14 days are administered postoperatively, avoiding "restarting the clock." (Consensus was reached.)
- In patients who completed therapy before cardiac valve surgery with no signs of active infection, including negative valve tissue cultures, no additional antibiotics are needed postoperatively. (Consensus was reached.)
- Duration of antimicrobial therapy needs to be individualized and informed by the endocarditis team.

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Treatment Regimen Change From NVE to PVE in Patients With Staphylococcal NVE Who Undergo Surgical Valve Intervention

There was a lack of consensus in the 2015 AHA scientific statement for patients with NVE who undergo surgical valve intervention as to whether the postoperative treatment regimen would remain as a native valve treatment regimen or change to a prosthetic valve treatment regimen.[3] A multicenter retrospective review by Katragadda et al[101] of patients with native valve *S aureus* IE who underwent surgical valve replacement or repair within the initial 6 weeks of antimicrobial therapy addresses this controversial topic. Patients received either monotherapy (native valve regimen) or combination (prosthetic valve regimen) therapy. The authors demonstrated by a propensity score–weighted analysis that patients treated with combination therapy had an increased risk of 6-month relapse, 1-year mortality, and treatment-related adverse events.[101] The addition of gentamicin with or without rifampin may cause more harm than benefit. However, this study had several limitations, including its retrospective, nonrandomized design and small sample size.

Key Take-Home Point

- Considering the increased risk of adverse drug events and poorer outcomes associated with combination therapy, continuation of monotherapy after cardiac valve surgery in patients with staphylococcal NVE appears reasonable. (Consensus was reached.)

Surgery for IE: Why, How, and When?

Surgery is a complement to antimicrobial treatment and indications generally reflect the presence of damaged valves causing heart failure, invasive infection with abscess formation, perforation or fistula formation, difficult-to-eradicate microorganisms, failure of antimicrobial therapy, prosthetic valve IE, or large (>1 cm) vegetations threatening immediate or recurrent embolization. The most recent iterations of international multisociety guidelines contain directionally consistent recommendations regarding the indications for and timing of surgical intervention in patients with IE.[7,16,54,102]

Of patients with IE undergoing surgery, ≈50% to 60% are operated on during the active phase of infection and before completion of a course of antibiotics at dedicated cardiac surgical programs, although multiinstitutional registry studies suggest a lower incidence of early operation. The decision to operate is made by a multidisciplinary team, with agreement among infectious disease, cardiology, and surgical members regarding indications, timing, and operability. Neurology's input is essential when central nervous system complications (eg, stroke, mycotic aneurysm) have occurred. Once the indication and need for surgery during the active phase of IE are established, early surgery within 48 to 72 hours is favored.

Surgery is performed to remove sources of embolism, drain abscesses, remove infected foreign material and necrotic tissue, and restore cardiac integrity and valve function. Intraoperative TEE guidance is essential for assessment of the extent of infection, extravalvular invasion, and adequacy of the surgery.

Indications for and the role of surgery in patients with right-sided IE, many of whom inject drugs, are less clear. Indications include persistent BSI, ongoing septic pulmonary embolization, pulmonary abscess formation after 7 days of antibiotics, or right-sided vegetations >2 cm. Repair of right-sided valves is preferred over replacement whenever feasible. Consideration of PMA to debulk vegetations may be considered in high surgical risk patients at institutions experienced in the technique (see section on PMA).[103]

Surgery in the context of active IE is inherently complicated and requires specialized expertise and experience. The surgeon needs to be prepared to handle disease that is found intraoperatively to be more advanced than diagnosed preoperatively. This is true for patients with either PVE or NVE.[104] Perioperative and early postoperative mortality rates vary across institutions (from 7% to 20%), suggesting opportunities to improve practices regarding patient selection, timing, perioperative management, and postoperative care. Specific technical details of the operation, choice of valve substitute when repair is not feasible, and type of conduit to use, when needed (eg, necessity to use allografts for aortic root replacement), are less important than attention to eradication of the infection and restoration of valve function. During surgery, if a CIED is present, the generator and leads are typically removed, with either intraoperative replacement using an epicardial system or postoperative implantation of a new device (see section on concurrent CIED infection and lead extraction).

Post-TAVR IE, a more recently recognized challenge, is related in part to the older age, higher burden of comorbidities, prevalent causative microorganisms (eg, enterococci, *S aureus*, coagulase-negative staphylococci), and higher surgical risk in this patient population.[26,105] Although TAVR valve explantation adds operative complexity and carries a risk of aortic root injury, patients with IE after TAVR are evaluated and managed similarly to other patients with PVE, including consideration of surgery when indicated and operative risk is acceptable.

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Key Take-Home Points

- Although endocarditis team members may agree on an indication for surgery, the final decision to operate rests with the cardiovascular surgeon.
- Surgery complements antimicrobial therapy by achieving source control, preventing embolism, and restoring cardiac integrity and function.
- When indicated, early surgery within 48 to 72 hours is favored, unless there has been a major stroke or cerebral hemorrhage (see Optimal Timing of Cardiovascular Surgery After Stroke).

Optimal Timing of Cardiovascular Surgery After Stroke

Acute stroke is the second most common complication associated with left-sided IE and, together with congestive heart failure, represents a leading cause of death due to IE.[106] A sentinel neurologic event is most likely to occur within the first 2 weeks after initiation of intravenous antibiotic treatment. Vegetation size, location, and mobility in addition to the duration of antimicrobial treatment determines the risk of an embolic event.[107] Patients with IE who have neurologic symptoms warrant collaboration with a neurologist and brain MRI to determine the severity of the stroke and whether it is embolic, ischemic, or hemorrhagic.

To date, no large, prospective studies have been conducted to guide decision-making regarding the timing of valve surgery for IE after a stroke. The primary concern when determining surgical timing is the risk of hemorrhagic conversion secondary to anticoagulation with cardiopulmonary bypass. In patients with small strokes on brain imaging without neurologic deficits, surgery can proceed without delay. For patients with hemorrhagic strokes or intraparenchymal bleeding with or without neurologic deficits, surgery is usually delayed at least 2 weeks, but preferably 4 weeks, depending on hemodynamic stability and neurologic evaluation. If a patient with embolic stroke without signs of bleeding on head imaging is hemodynamically unstable due to acute-onset heart failure, uncontrolled sepsis, or cardiac abscess, or is at a high risk for recurrent embolism, urgent (<72 hours) or emergent surgery should be considered.

The criteria to determine whether patients are at a high risk for recurrent embolic events include large or mobile vegetations >10 mm, mitral valve involvement (especially anterior leaflet involvement), and staphylococcal IE. It is imperative to obtain CT or CT angiography, or MRI or magnetic resonance angiography, to rule out myotic aneurysm before proceeding with cardiothoracic surgery. If neurologic imaging demonstrates an embolic stroke with intracranial hemorrhage, early surgery may be considered in select circumstances when there is a high risk of recurrent embolism, hemodynamic compromise, sepsis, or cardiac abscess. If those findings are not present, it is appropriate to defer surgery for 3 to 4 weeks.

Timing of surgery is to be determined by the endocarditis team with input from neurology and neurosurgery experts. In addition to considerations regarding cardiac surgical timing, neurologic intervention with mechanical thrombectomy (MT) is being examined as a strategy to restore patients to their previous level of functioning.

Key Take-Home Points

- Cardiac surgery is not appropriate when patients are comatose.
- Hemorrhagic and major ischemic strokes usually delay cardiac surgery at least 2 weeks, but preferably 4 weeks.
- Timing of surgery is to be determined by the endocarditis team with input from neurology and neurosurgery experts.

Role of MT in Patients With Ischemic Stroke

Data are limited regarding the efficacy of MT for ischemic strokes secondary to IE. In a previous study, patients with an acute ischemic stroke secondary to IE who were treated with intravenous thrombolysis had a 20% increase in mortality due to hemorrhagic complications.[108] MT does not appear to result in hemorrhagic complications. MT has been noted to increase vasospasm and cause endothelial damage at the site of the infective embolus. A recent systematic review suggested that although postprocedure hemorrhage rates were high in patients who received intravenous thrombolysis compared with those who underwent MT, neurologic prognosis and mortality rates were similar between the groups.[109] MT continues to be a feasible treatment option for cardiac patients with ischemic stroke secondary to a large vessel occlusion.[110]

Key Take-Home Point

- More data are needed to better define the role of MT for ischemic strokes secondary to IE.

CIED Extraction in Valvular IE

The presence of a CIED increases the risk of IE and is included as a minor criterion in the 2023 Duke–ISCVID criteria.[11] In the setting of bacteremia, there can be seeding of both intracardiac device leads and the cardiac valves. In the 2024 AHA scientific statement on CIED infections, CIED IE requires complete device extraction due to high relapse rates when the device is retained or partially removed.[5] In a recent Danish nationwide study of nonsurgically managed IE, CIED removal was associated with

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significantly lower 6-month rates of IE readmission, recurrent bacteremia, and mortality compared with those without CIED removal.[111] Complete device extraction may be performed for all patients with valvular IE without evidence of lead vegetations or pocket generator findings.[5]

Key Take-Home Points

- Presence of a CIED increases the risk of IE.
- In valvular IE without evidence of CIED lead vegetations or pocket generator findings, complete device extraction is preferred.

PMA in IE

PMA involves imaging-guided catheter-based removal of vegetations.[10] The core tenet of PMA use in IE is achievement of source control through debulking of vegetations, often with recurrent pulmonary emboli, sepsis, and sustained BSI despite appropriate antimicrobial treatment. PMA is a novel procedure for select patients with right-sided IE; its use has escalated and is currently being evaluated for left-sided IE. PMA is an option for patients deemed not to be surgical candidates for various reasons. Most patients undergoing PMA are PWID or have a CIED infection with suspected lead involvement.

Because most of these infections are due to *S aureus*, a pathogen well-recognized for its virulence, poorer outcomes have been observed. Patients with tricuspid valve or lead vegetations >2 cm have a poorer prognosis and are potential candidates for PMA.[112,113] There are no RCT data to determine the role of PMA in patient management.[103] The 2023 ESC guidelines included a Class IIb with a Level of Evidence C recommendation that PMA may be considered in select patients with right intra-atrial septic masses and high surgical risk.[16] Expertise in PMA is required, but not widely available.

Key Take-Home Points

- PMA has been used in select patients with tricuspid valve or CIED lead–related IE and high surgical risk when medical therapy has failed.
- The efficacy and safety of PMA require further study.

PROLONGED ANTIMICROBIAL SUPPRESSION

Indications for Long-Term Antibiotic Suppression for IE

A consequence of medical progress is that an increasing proportion of IE cases affect an older population[114,115] and those with prosthetic valves[116] or intracardiac devices.[117] Long-term antibiotic suppression (suppressive antimicrobial therapy [SAT]) may be a valuable therapeutic approach in select cases, such high risk for IE relapse, poor surgical candidacy with foreign material not extracted despite an indication, limited life expectancy, patient preference, or other scenarios following an individualized risk–benefit approach. SAT in IE is a secondary prevention approach designed to decrease relapse risk through long-term antimicrobial treatment following the initial curative and continuation phases of antibiotic therapy. When pursued, pathogen-directed SAT therapy with an oral agent is preferred. High-quality evidence on SAT in IE is limited to a small number of nonrandomized clinical studies, and there are no data available on the optimal dosage and duration of SAT.

The most widely studied indication for SAT is in patients with a surgical indication in whom surgery is not performed. In this scenario, SAT can reduce the relapse rate to between 7% and 12.5%,[118–121] underscoring that breakthrough infections represent a substantial risk despite SAT. In a case series of IE due to Gram-positive cocci, both oral antimicrobial options and intravenous dalbavancin (500 mg weekly or 1000 mg biweekly regimens) have been effective.[120–122]

SAT in patients with CIED IE or partial CIED removal is discussed in-depth in the 2024 AHA CIED infection scientific statement.[5] There are no data on the residual risk associated with abandoned or incomplete lead extraction in patients with CIED infection and concomitant valvular endocarditis involving retained leads. In this setting, ^{18}F-FDG-PET/CT may be useful to evaluate for residual infection and to help determine whether SAT therapy is warranted after completion of initial curative-phase antibiotic therapy.[123] Patients with fungal PVE may also benefit from SAT. Mortality rates in fungal PVE are high (>50%), and relapse rates may be as high as 36%.[124,125] Long-term suppressive treatment with oral azoles is used, sometimes lifelong, to reduce relapse rates. Mycobacterial IE is rare and associated with prolonged antimicrobial therapy and high mortality rates, especially without surgical intervention.[126] Duration of therapy for mycobacterial IE is undefined, with consideration for lifelong SAT especially in cases without source control.[126]

SAT is not a substitute for surgical intervention in the absence of contraindications to surgery. Duration of SAT is individualized and can be monitored by ^{18}F-FDG-PET/CT and ongoing evaluation of drug tolerability. Patients receiving SAT require early follow-up to evaluate for infection relapse and adverse drug events. BSI persisting for a short duration (≈1 or 2 days) after surgical source control is not an indication for SAT in the absence of another more compelling indication. RCTs and longer prospective follow-up studies will be needed to establish SAT efficiency, refine indications, determine adverse drug event rates, and establish best practices on recommended durations.

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Key Take-Home Points

- SAT is reasonable for patients with established PVE or CIED IE and an indication for surgery but who are not offered surgery due to unacceptably high operative risk related to age, fragility, or comorbidities.
- SAT is often considered in fungal and mycobacterial IE, especially in cases without source control.

FUTURE DIRECTIONS

There is a striking paucity of data in multiple areas of the diagnosis and management of IE to guide optimal patient care. Recognizing that RCTs in IE are difficult to perform, evidence derived from other types of studies (eg, registries, natural history studies, emulated trials) is sorely needed. The writing group also advocates for the creation of a consortium of medical centers with IE interest and expertise to provide more evidence-based recommendations (ie, an international IE clinical trials platform). Future studies evaluating real-world use of POT, long-acting injectable lipoglycopeptides, and PMA in select patients with IE to enhance treatment options are needed.

Both laboratory and radiologic advances have been revolutionary in the diagnosis and management of IE. Advanced microbiologic diagnostics will enhance earlier initiation of appropriate antimicrobial therapy. Therapeutic drug monitoring as exemplified by the POET trial has become increasingly available and this trend is likely to continue.

Novel radiopharmaceuticals are expected to improve the diagnostic capabilities of imaging in the diagnosis and monitoring of patients with IE. The optimal timing of PET/CT in the diagnosis of IE and to decide when to stop SAT is of interest and investigations are needed to enhance the sensitivity of this screening, which is susceptible to cumulative antimicrobial treatment.

The complexities of IE in PWID have been recognized[1] and we expect more medical centers will continue to develop unique management strategies so that substance use, psychiatric diagnoses, and social issues, in particular housing instability and barriers to transportation, are addressed. Screening for coinfection, such as HIV, hepatitis B and C, and sexually transmitted infections, is also needed; more information is available in a previous AHA scientific statement on IE in PWID.[1] Stigma contributes to disparities in health outcomes across sex, racial, and ethnic groups. Addressing these barriers is essential in both current care and future management strategies. Changes in traditional educational training programs are needed to accommodate residents and fellows who have a career interest in managing patients with substance use disorders, psychiatric diagnoses, or socioeconomic challenges. Such changes may include formal clinical rotations in psychiatry and addiction medicine, as well as collaboration with social workers to better understand the unique needs of PWID.

Artificial intelligence is being developed for all aspects of clinical research that ultimately address patient care. Results of large language models are being used in individual patient care. The writing group recognizes the importance of the clinical use of artificial intelligence today and in the future. The use of artificial intelligence is a potentially invaluable tool for the diagnosis of IE based on the electronic medical record, image interpretation, and risk stratification of people with increased risk of IE. The accuracy of information provided by large language models requires assessment by specialists in the field, and preferably by an endocarditis team, for interpretation of results generated. Human evaluation remains essential in scrutiny of these results.

Funding for these future advances will require both traditional and novel avenues to secure funding, especially for low- and middle-income countries, and will require local, regional, national, and international engagement.

ARTICLE INFORMATION

The American Heart Association makes every effort to avoid any actual or potential conflicts of interest that may arise as a result of an outside relationship or a personal, professional, or business interest of a member of the writing panel. Specifically, all members of the writing group are required to complete and submit a Disclosure Questionnaire showing all such relationships that might be perceived as real or potential conflicts of interest.

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*Modest.

†Significant.

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†Significant.

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